Final Exam 2022 Objectives of Zoo 201
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1. The **Chinese liver fluke**, Clonorchis sinensis, is a common parasite of humans in Asia, where over **30 million** people are infected.
2. Arachnids use ***coxal glands and/or Malpighian tubules*** fore xcreting nitrogenous wastes
3. *Diphyllobothrium latum,* is relatively common in the northern parts of ***North America*** and in the Great Lakes area of ***the United States.***
1. Echinoids move by using spines for pushing against the substrate ***and tube feet*** for pulling.
2. The free-living and parasitic ways of life formed in the Cambrian period, ***600 million*** years ago.
3. In bivalves, blood flows from the heart to tissue sinuses, nephridia, gills, and back to the **heart.**
4. In some bivalves, the separate aorta delivers blood directly to the ***mantle.***
5. Some surface-dwelling bivalves attach to the substrate either by proteinaceous strands ***called byssal threads***
6. **Malpighian tubules,**absorb waste materials from the blood and empty them into the gut tract.
7. Nematodes Some ***five billion*** may be in every acre (4,046 square meters) of fertile garden soil.
8. Multicellular life has been a part of the earth's history for approximately ***550 million*** years.
9. The major excretory product of arachnids is ***uric acid.***
10. Large tissue spaces, or sinuses, are derived from the blastocoel and are called the **hemocoel.**
11. A statocyst is a pit like invagination of the exoskeleton that contains setae and a group of cemented sand grains called a **statolith.**
12. Crayfish have a lens system consisting of 25 to 14,000 individual receptors called ***ommatidia.***
13. The excretory organs of crayfish are called **antennal glands** (green glands) because they are at the bases of the second antennae and are green in living crayfish.
14. In other crustaceans, they are called **maxillary glands** because they are at the bases of the second maxillae.
15. Many other crustaceans have a planktonic, free-swimming larva called a **nauplius.**

16. Crabs and their relatives have a second larval stage called a **zoea**.
17. When all adult features are present, except sexual maturity, the immature is called the **post larva.**
18. A **nematocystis** a fluid-filled, intracellular capsule enclosing a coiled hollow tube.
19. The cnidocyte has a modified cilium, called a **cnidocil.**
20. Male millipedes transfer sperm to female millipedes with modified trunk appendages, called **gonopods, or in spermatophores**
21. Insect **thermoregulationis** the process whereby insects maintain body temperatures within certain boundaries.
22. **Heterothermy** ("hetero", different) occurs inanimals which are able to switch from endothermy to ectothermy.
23. **Pheromones** are chemicals an animal releases that cause behavioral or physiological changes in another member of the same species
24. Each kind of individual in an insect colony is called **a caste.**
25. Reproductive males inseminate the queen (s) and are called **kings or drones.**
26. A **pheromone** that the queen releases controls the honeybee caste system.
27. The scale insect, ***Iceryapurchasi***, was introduced into California in the 1860s.
28. An insect that transmits a disease is known as a **vector**, and the disease is referred to as an avector-borne disease.
29. The echinoderm skeleton consists of a series of calcium carbonate plates called **ossicles**.
30. In sea urchins and sand dollars, these inter locking plates form an enclosure, almost like a shell, called a **test.**
31. The water-vascular system of echinoderms is a series of water-filled canals, and their extensions are **called tube feet**.
32. Other terms in echinoderms sometimes used to refer to the water vascular system are **"ambulacral system"and "aquiferous system".**
33. **A hemal system** consists of strands of tissue that encircle an echinoderm near the ring canal of the water-vascular system.
34. In some sea stars, the aboral surface has numerous pincherlike structures called **pedicellariae**.
35. Apedicellaria(plural:pedicellariae) is a small wrench-or claw-shaped appendage with movable jaws, called **valves.**
36. The larger, oral stomach, sometimes called the **cardiac stomach**, receives ingested food.
37. It joins the smaller, aboral stomach, sometimes called **the pyloric stomach**.
38. The aboral (pyloric) stomach gives rise to ducts that connect to secretory and absorptive structures called **pyloric cecae**.

39. After gastrulation, in echinoderms bands of cilia differentiate, and a bilaterally symmetrical larva, called a **bipinnaria larva**, forms.
40. **Ophiuroidea** is Closely related to starfish.
41. Ammonia is the primary nitrogenous waste product, and it is lost by diffusion across tube feet and membranous sacs, **called bursae**,
42. A larval stage, called an ***ophiopluteus,*** is planktonic.
43. The **pedicellariae** of sea urchins have either **two or three jaws** and connect to the body wall by a relatively long stalk.
44. A chewing apparatus, called **Aristotle's lantern**, can be projected from the mouth.
45. The digestive tract forms a U-shaped gut in the **calyx.**
46. There is no larval stage or metamorphosis in **Chaetognaths.**
47. Sea cucumbers can evert tubules of the respiratory tree, called **Cuverian tubules.**
48. A **pheromone** that the queen releases controls the honeybee caste system
49. The body of an insect is divided into three tagmata: **head, thorax, andabdomen**.
50. An **insect growth regulator (IGR)** is a substance (chemical) that inhibits the life cycle of an insect.
51. Grasshopper tympanal organs can detect sounds in the range of **1,000 to50,000 cps.**
52. The human ear can detect sounds between **20 and 20,000 cps**.
53. **Thermoregulation** is a requirement for flying insects.
54. Arrow worms are active **predators.**
55. Arrow worms are active **swimmers**.
56. Most Chaetognaths are planktonic, **0.5 to 12 cm** in length,
57. Entoprocts have a small body cavity that most zoologists consider a **pseudocoelom.**
58. Most entoprocts are **microscopic.**
59. Cycliophores can reproduce asexually by producing a larva stage, called the **Pandora** that is released and settles on the same host.
60. In Cycliophores **Protonephridia** are used in excretion.
61. The asexual feeding stage of symbion is about **0.3 mm**tall.
62. The body wall contains gland cells that eventually secrete the **chitinous tube.**
63. The **gonads** are in the coelom.
64. A flap of tissue called the **epistome** covers the mouth.
65. The Phoronids consist of about a dozen marine species divided between two genera: **Phoronis and Phoronopsis.**

66. **Statoblasts** are asexually produced and accumulate in the **metacoel.**
67. A statoblast is a hard, resistant capsule containing a mass of **germinativecells.**
68. Some freshwater ectoprocts produce a dormant stage called a **statoblast.**
69. In others, simple gonads produce both sperm and eggs in the same **autozooid**.
70. Each colony can contain about **2 million zooids.**
71. Most ectoprocts are **monoecious**.
72. Echinoids are **dioecious.**
73. Sea cucumbers are **dioecious.**
74. Ectoprocts grow by**budding**
75. All species are aquatic (both freshwater and marine) and less than **1.5 mm** in length.
76. Brachiopods are commonly called **lampshells** because they resemble **ancientRoman oil lamps.**
77. An anus always open outside of the **lophophore.**
78. **Metamorphosis** occurs after larvae attach to the substrate.
79. The unattached end of a sea lily is called the **crown.**
80. Sea lilies attach permanently to their substrate by a **stalk**.
81. Members of the class **Crinoidea** include the **sea lilies and the feather stars.**
82. Sea cucumbers have **no true brain.**
83. Most sea cucumbers ingest particulate organic matter using their **tentacles.**
84. The class Holothuroidea has approximately **1,500 species**, whose members are commonly called **sea cucumbers**.
85. A chewing apparatus, **called Aristotle's lantern**, can be projected from the mouth.